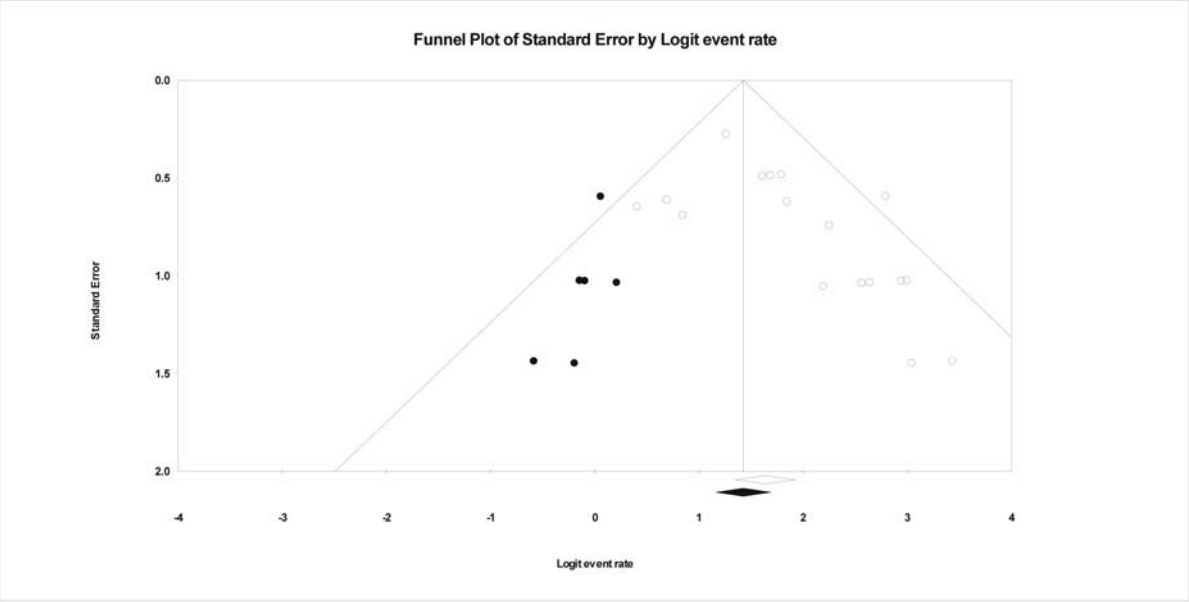
Supplementary material

Supplementary Fig. 1 Adjusted funnel plot using "trim and fill" test with actual and imputed studies. Visible asymmetry demonstrated based on clinical success using EVAC therapy. The hollow circle represents actual study, the hollow diamond represents actual effect size, solid circles represent imputed studies, and the solid diamond represents adjusted effect size.



Supplementary material

Supplementary Table 1 Search strategy using EMBASE.

No.	Query	Results
#1	'endoluminal vacuum*' OR 'endoscopic vacuum*' OR 'endo vac*' OR endovac* OR 'e vac' OR 'endoluminal negative pressure*' OR 'vacuum assisted closure*' OR 'vacuum therapy device*' OR 'endo sponge*' OR endosponge* OR 'negative pressure therapy' OR 'vacuum assisted closure'/exp OR 'vacuum therapy device'/exp	8211
#2	esophag* OR oesophag* OR 'esophagus perforation'/exp OR 'esophagus fistula'/exp OR 'esophagus'/exp	327266
#3	leak* OR fistula* OR perforat* OR 'esophagus perforation'/exp OR 'esophagus fistula'/exp OR 'anastomosis leakage'/exp	480865
#4	#2 AND #3	39425
#5	#1 AND #4	302
#6	#5 NOT ('animal experiment'/de OR 'animal model'/de OR 'animal tissue'/de OR 'case report'/de OR 'meta analysis'/de OR 'nonhuman'/de OR 'porcine model'/de OR 'practice guideline'/de OR 'systematic review'/de)	171
#7	#6 NOT ('conference review'/it OR 'editorial'/it OR 'review'/it)	141

Supplementary material

Supplementary Table 2 Types of sponge and vacuum systems used in different studies.

Study	Year	Sponge type	Pressure	Interval change
Berlth et al.	2018	open-pore polyurethane foam	125 mmHg, 3-5 day change	Every 3-5 days
Bludau et al.	2018	polyurethane foam drainage tube (e.g., EsoSponge, Braun)	100-125, 3-4 days	Every 3-4 days
Brangewitz et al.	2013	polyurethane foam sponge (pore size 400–600 µm; KCl, Wiesbaden Germany	125 mmHg, twice a week removal	twice a week
Heits et al.	2014	Endo-Sponge system (B. Braun, Melsungen, Germany)	75-200 mmHg, every 2 days and twice a week when clean woulnd and granulation tissue detected	Every 2 days to twice a week when clean wound and granulation tissue detected
Jeon et al.	2019	polyurethane sponge (KCl, San Antonio, TX)	125 mmHg, Q3-4 days	Every 3-4 days
Jung et al	2020	Esosponge Braun B Melsungen, Germany	20-50 mmHg Q 3-4 days	Every 3-4 days
Kuehn et al.	2015	polyurethane sponges of the BEndosponge system (B. Braun, Melsungen, Germany	125 mmHg Q2-4 days	Every 2-4 days
Laukoetter et al.	2016	open-pore polyurethane sponge (e.g., VivanoMed Foam, Paul Hartmann AG, Heidenheim, Germany)	100-120, 3-5 days	Every 3-5 days
Loske (1) et al.	2011	Open-pore polyurethane sponge (V.A.C. GranuFoam, KCl Inc. San Antonio, USA; Endo-SPONGE, B. Braun, Melsungen, Germany)	125 mmHg, 1-8 days	Every 1-8 days
Loske (2) et al.	2015	open pore polyurethane foam (V.A.C. GranuFoam, Kinectic Concepts, Inc., San Antonio, USA; Endo-SPONGE, B. Braun Melsungen AG, Melsungen, Germany; SuprasorbCNP Wundschaum, Lohmann& Rauscher GmbH& Co. KG, Neuwied, Germany	125 mmHg, Q 2-5 days	Every 2-5 days
Mencio et al.	2018	granulofoam (KCl/Acelity, San Antonio, TX)	175 mmHg, 4.8 days	Every 4.8 days in average
Mennigen et al.	2015	Polyurethane foam sponge (e.g., V.A.C. GranuFoam, pore size 400–600 µm; KCl—KineticConcepts, Inc., TX, USA, and Wiesbaden, Germany)	100-125, 3-5d	Every 3-5 days
Min et al.	2019	Polyurethane sponge	100mmHg Q3-4 d	Every 3-4 days
Moschler et al.	2015	EndoSponge system (B. Braun Melsungen Ltd., Melsungen, Germany)	100-125 Q3-5 days	Every 3-5 days
Noh et al.	2018	VAC kit (CuraVAC; CGBio Inc., Hwaseong, Korea) that has a medical-grade polyurethane sponge	80-125, every 1-2 weeks (7-14 days)	Every 1-2 weeks (7-14 days)
Ooi et al.	2018	standard VAC dressing kit (polyurethane sponge)	100-125, 2-4 days	Every 2-4 days
Pournaras et al.	2018	V.A.C.– GranuFoamTM Small Dressing Kit, KCl	125 mmHg, 2-3 days	Every 2-3 days
Schneiwind et al.	2013	EndoSponge, B. Braun, Melsungen, Germany	70-80 mmHg, twice a week	Twice a week

Supplementary material

Supplementary Table 3 Definition and rate of technical and clinical success in each study. n, number of patients; NR, not reported.

Study	Total patients (n)	Definition of technical success	Technical success, n (%)	Definition of clinical success	Clinical success, n (%)
Berlth et al.	35	Uneventful and successful application of the endoscopic treatment	29 (82.9%)	No leak or fistula at the end of treatment. No treatment change or death before healing. No need for surgery, stent, or referral.	30 (85.7%)
Bludau et al.	77	Successful application of the endoscopic treatment	72 (93.5%)	No leak or fistula at the end of treatment. No treatment change or death before healing.	60 (77.9%)
Brangewitz et al.	32	Successful application of the endoscopic treatment	32 (100.0%)	No leak or fistula at the end of treatment confirmed with endoscopy and radiology with no recurrence in hospital discharge follow up.	27 (84.4%)
Heits et al.	10	Survival and successful application of the endoscopic treatment	9 (90.0%)	Survival to reach macroscopic establishment of mucosal healing	9 (90.0%)
Jeon et al.	22	No technical difficulty in application of endoscopic treatment	21 (95.5%)	No leak or fistula at the end of treatment	19 (90.5%)
Jung et al.	30	Survival and successful application of the endoscopic treatment	29 (96.7%)	Survival to reach mucosal healing	25 (83.3%)
Kuehn et al.	21	Survival and successful application of the endoscopic treatment	19 (90.5%)	No leak or fistula at the end of treatment	19 (90.5%)
Laukoetter et al.	52	Successful application of the endoscopic treatment	52 (100.0%)	No leak or fistula at the end of treatment confirmed with endoscopy and radiology	49 (94.2%)
Loske (1) et al.	14	Successful application of the endoscopic treatment	14 (100.0%)	No leak or fistula at the end of treatment	13 (92.9%)
Loske (2) et al.	10	Successful application of the endoscopic treatment	10 (100.0%)	No leak or fistula at the end of treatment	10 (100.0%)
Mencio et al.	15	Successful application of the endoscopic treatment	15 (100.0%)	No leak or fistula at the end of treatment	15 (100.0%)
Mennigen et al.	15	Successful application of the endoscopic treatment	15 (100.0%)	No leak or fistula at the end of treatment confirmed with endoscopy and radiology	14 (93.3%)
Min et al.	20	Survival and successful application of the endoscopic treatment	20 (100.0%)	Survival to reach mucosal healing	19 (95.0%)
Moschler et al.	10	Successful application of the endoscopic treatment	9 (90.0%)	Survival to reach mucosal healing	7 (70.0%)
Noh et al.	12	Successful application of the endoscopic treatment	12 (100.0%)	No leak or fistula at the end of treatment confirmed with endoscopy and radiology. Able to advance diet.	8 (66.7%)
Ooi et al.	10	Successful application of the endoscopic treatment	9 (90.0%)	No leak or fistula at the end of treatment confirmed with endoscopy and radiology	6 (60.0%)
Pournaras et al.	21	Survival and successful application of the endoscopic treatment	20 (95.2%)	Survival to reach mucosal healing	20 (95.2%)
Schneiwind et al.	17	NR	NR	NR	NR

Supplementary material

Supplementary Table 4 Newcastle Ottawa Scale for risk of bias in individual studies.

Study	Selection				Comparability		Outcome		Quality score
	Representativeness of the exposed cohort	Selection of non-exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at the start of the study	Comparability of the cohorts on the basis of design or analysis	Assessment of outcome	Was follow up long enough for outcomes to occur	Adequacy of follow up cohorts	
Berlth et al.	1	1	1	1	1	1	1	1	8
Bludau et al.	1	0	1	1	0	1	1	1	6
Brangewitz et al.	1	1	1	1	1	1	1	1	8
Heits et al.	1	0	1	1	0	1	1	1	6
Jeon et al.	1	0	1	1	0	1	1	1	6
Jung et al	1	0	1	1	0	1	1	1	6
Kuehn et al.	1	0	1	1	0	1	1	1	6
Laukoetter et al.	1	0	1	1	0	1	1	1	6
Loske (1) et al.	1	0	1	1	0	1	1	1	6
Loske (2) et al.	1	0	1	1	0	1	1	1	6
Mencio et al.	1	0	1	1	0	1	1	1	6
Mennigen et al.	1	1	1	1	1	1	1	1	8
Min et al.	1	0	1	1	0	1	1	1	6
Moschler et al.	1	0	1	1	0	1	1	1	6
Noh et al.	1	0	1	1	0	1	1	1	6
Ooi et al.	1	0	1	1	0	1	1	1	6
Pournaras et al.	1	0	1	1	0	1	1	1	6
Schneiwind et al.	1	1	1	1	1	0	1	1	7

Supplementary material

Supplementary Material (References for included studies):

1. Kuehn F, Schiffmann L, Janisch F, Schwandner F, Alsfasser G, Gock M, et al. Surgical Endoscopic Vacuum Therapy for Defects of the Upper Gastrointestinal Tract. J Gastrointest Surg. 2016;20:237-43
2. Ooi G, Burton P, Packiyanathan A, et al. Indications and efficacy of endoscopic vacuum-assisted closure therapy for upper gastrointestinal perforations. ANZ J Surg. 2018;88:E257-E263. doi:10.1111/ans.13837
3. Berlth F, Bludau M, Plum PS, et al. Self-Expanding Metal Stents Versus Endoscopic Vacuum Therapy in Anastomotic Leak Treatment After Oncologic Gastroesophageal Surgery. J Gastrointest Surg. 2019;23:67-75. doi:10.1007/s11605-018-4000-x
4. Bludau M, Fuchs HF, Herbold T, et al. Results of endoscopic vacuum-assisted closure device for treatment of upper GI leaks. Surg Endosc. 2018;32:1906-1914. doi:10.1007/s00464-017-5883-4
5. Brangewitz M, Voigtländer T, Helfritz FA, et al. Endoscopic closure of esophageal intrathoracic leaks: stent versus endoscopic vacuum-assisted closure, a retrospective analysis. Endoscopy. 2013;45:433-438. doi:10.1055/s-0032-1326435
6. Jeon JH, Jang HJ, Han JE, et al. Endoscopic Vacuum Therapy in the Management of Postoperative Leakage After Esophagectomy. World J Surg. 2020;44:179-185. doi:10.1007/s00268-019-05228-z.
7. Jung CFM, Müller-Dornieden A, Gaedcke J, et al. Impact of Endoscopic Vacuum Therapy with Low Negative Pressure for Esophageal Perforations and Postoperative Anastomotic Esophageal Leaks. Digestion. 2020;1-11. doi:10.1159/000506101
8. Laukoetter MG, Mennigen R, Neumann PA, et al. Successful closure of defects in the upper gastrointestinal tract by endoscopic vacuum therapy (EVT): a prospective cohort study. Surg Endosc. 2017;31:2687-2696. doi:10.1007/s00464-016-5265-3.
9. Loske G, Schorsch T, Müller C. Intraluminal and intracavitary vacuum therapy for esophageal leakage: a new endoscopic minimally invasive approach. Endoscopy. 2011;43:540-544. doi:10.1055/s-0030-1256345.
10. Loske G, Schorsch T, Dahm C, Martens E, Müller C. Iatrogenic perforation of esophagus successfully treated with Endoscopic Vacuum Therapy (EVT). Endosc Int Open. 2015;3:E547-E551. doi:10.1055/s-0034-1392566.
11. Mencio MA, Ontiveros E, Burdick JS, Leeds SG. Use of a novel technique to manage gastrointestinal leaks with endoluminal negative pressure: a single institution experience. Surg Endosc. 2018;32:3349-3356. doi:10.1007/s00464-018-6055-x.
12. Mennigen R, Harting C, Lindner K, et al. Comparison of Endoscopic Vacuum Therapy Versus Stent for Anastomotic Leak After Esophagectomy. J Gastrointest Surg. 2015;19:1229-1235. doi:10.1007/s11605-015-2847-7.
13. Möschler O, Nies C, Mueller MK. Endoscopic vacuum therapy for esophageal perforations and leakages. Endosc Int Open. 2015;3:E554-E558. doi:10.1055/s-0034-1392568.
14. Noh SM, Ahn JY, Lee JH, et al. Endoscopic Vacuum-Assisted Closure Therapy in Patients with Anastomotic Leakage after Esophagectomy: A Single-Center Experience. Gastroenterol Res Pract. 2018;2018:1697968. doi:10.1155/2018/1697968.
15. Heits N, Stapel L, Reichert B, et al. Endoscopic endoluminal vacuum therapy in esophageal perforation. Ann Thorac Surg. 2014;97:1029-1035. doi:10.1016/j.athoracsur.2013.11.014.

Supplementary material

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1, title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3, 4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7 supplementary table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7, 8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8, 9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8, 9

Supplementary material

Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

Page 1 of 2

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10, Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11, supplementary table 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Table 2
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10, 11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Supplementary table 2, Supplemenary Figure 1
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13, 14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.

Supplementary material

MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	5
2	Hypothesis statement	6
3	Description of study outcome(s)	8
4	Type of exposure or intervention used	8
5	Type of study designs used	8
6	Study population	8
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	7, title page
8	Search strategy, including time period included in the synthesis and key words	7
9	Effort to include all available studies, including contact with authors	7, 8
10	Databases and registries searched	7
11	Search software used, name and version, including special features used (eg, explosion)	NA
12	Use of hand searching (eg, reference lists of obtained articles)	7
13	List of citations located and those excluded, including justification	Supplementary material
14	Method of addressing articles published in languages other than English	8
15	Method of handling abstracts and unpublished studies	8
16	Description of any contact with authors	NA
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	8

Supplementary material

18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	8, 9
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	8, 9
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	NA
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	9
22	Assessment of heterogeneity	9
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	9
24	Provision of appropriate tables and graphics	9, Tables, Figures
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	11, Table 2, Figure 2-4
26	Table giving descriptive information for each study included	Table 1
27	Results of sensitivity testing (eg, subgroup analysis)	11
28	Indication of statistical uncertainty of findings	11

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	11, Supplementary Figure 1
30	Justification for exclusion (eg, exclusion of non-English language citations)	Figure 1
31	Assessment of quality of included studies	11, supplementary table 2
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	12, 13
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	13, 14
34	Guidelines for future research	14
35	Disclosure of funding source	Title Page

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.